A Kakeya Set Miniature: Tubes in the Unit Cube

Test Fixture

A *Kakeya set* in $\mathbb{R}^3$ contains a unit line segment in every direction. We study the Besicovitch–Kakeya conjecture at finite scale: a family $\mathbb{T}$ of δ-tubes pointing in δ-separated directions satisfies the Wolff axioms, and we measure the volume of its union on a voxel grid.

1 Setup

Take $\delta = 2^{-4}$ and $N = 96$ tubes through directions sampled on the Fibonacci hemisphere. The Kakeya maximal estimate we probe numerically is

$$\Big|\bigcup_{T\in\mathbb{T}} Y(T)\Big| \;\geq\; \delta^{\varepsilon}\lambda^{K}\sum_{T\in\mathbb{T}}|T|.$$

Dragging the packing parameter from separated to sticky, all tubes converge without changing directions; the multiplicity of the central voxel grows toward N.

2 Observation

At finite resolution the union volume decreases as the family packs toward the Kakeya configuration, in line with the sticky Kakeya phenomenon.